Offset Correction Methods: Laser Trim, e-Trim™, and Chopper

Ying Zhou, Art Kay

TEXAS INSTRUMENTS

Introduction

Amplifier input offset voltage is often the key parameter for design accuracy, and there are different circuit techniques used to minimize it. Knowing the various trimming and offset correction methods may be productive when selecting an amplifier for a particular application.

Laser trim: wafer-level trim

Each amplifier is composed of tens to thousands of transistors, resistors, and capacitors. The device input offset error is caused by the mismatch (or variations) of the input transistors in each amplifier. Figure 1 shows a photo of an op amp die. The highlighted areas show the thin film resistors. During the wafer-level test process, a laser burns away sections of resistive material, thus adjusting its value by increasing the total resistance of the die.

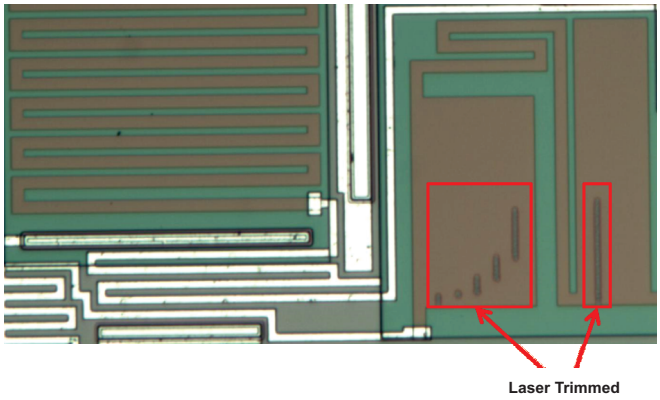


Figure 1. Laser Trimming on Thin-Film Resistors

This trimming can actually be done while the device is in operation under the laser; that is, we can monitor the offset voltage and then trim the resistor until the offset is nulled to zero. Wafer-level trimming may produce very accurate results. However, after the wafer is sawn into individual dies and packaged, any physical stress to a given die can produce some shifts in the offset voltage.

Laser trim is a practical way to trim ICs fabricated on a bipolar wafer process. It is used widely not only for op amps, but also in difference and instrumentation amplifiers to improve the resistor matching that is critical in minimizing gain error while optimizing common-mode rejection ratio (CMRR) performance. The INA826S is an instrumentation amplifier that uses laser trim to achieve high DC accuracies and low quiescent current: 150-µV offset, 2-µV/°C drift, and 250-µA quiescent current. With the additional shutdown function and small (3-mm × 3-mm) DFN package, this amplifier is suitable for power-sensitive applications such as circuit breakers, portable medical instrumentation, and test equipment.

e-Trim™: package-level trim

Another method used to achieve low offset voltage and drift is e-Trim, which is TI's patented trimming architecture implemented after device packaging. Correction current sources internal to the devices are adjusted during the final package-level manufacturing test. Once the trimming is completed, the gateway to trim circuitry is closed, so the trim control circuit is disabled and the adjustments become permanent. Figure 2 illustrates the e-Trim method.

This polysilicon, fuse-blowing technique does not require extra pins or test pads and provides substantial improved performance over wafer-level trim. This approach also avoids the parameter shifts that typically occur as a result of package stress, producing the excellent precision only possible with zero-drift amplifiers and allowing miniature package sizes.

Copyright © 2017, Texas Instruments Incorporated

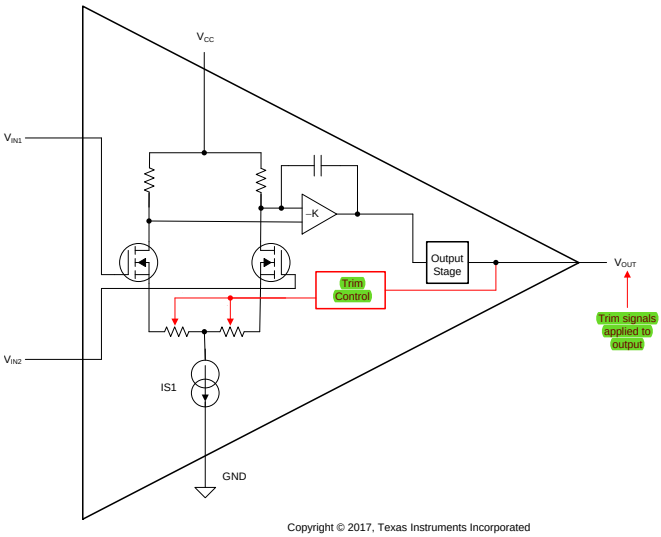


Figure 2. Package-Level e-Trim

The OPA2191 is the newest addition to the e-Trim amplifier family, featuring 25-µV (max) offset voltage, 0.8-µV/°C (max) drift, and low quiescent current of 200 µA per amplifier.

Chopper: dynamic correction

Although it is not a trim method as such, a third method to correct offset and drift is with a zero-drift amplifier. The chopper is a common type of zero-drift amplifier that uses an internal dynamic calibration method to minimize the offset voltage. Figure 3 shows the diagram of a typical chopper architecture. The input and output of the first trans-conductance stage has a set of switches that swaps the polarity of the input signal once every calibration cycle. The drift over time and temperature is averaged to zero. This characteristic also eliminates the 1/f noise that occurs in conventional amplifiers at very low frequencies. These features make chopper amplifiers very useful in DC (or low-frequency) signal conditioning applications such as weigh scales, strain gauges, and temperature measurement. The OPA2187 is an example of a chopper with 10-µV (max) offset voltage, 0.05-µV/°C (max) drift, and 0.4-μV_{PP} low-frequency noise (0.1 Hz to 10 Hz), all with only 145-µA (max) quiescent current per amplifier.

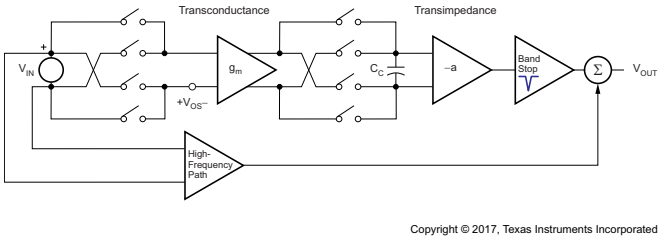


Figure 3. Typical Diagram of Chopper Amplifier

Chopper amplifiers, however, may insert an additional noise component caused by charge injection because of the switch closing and opening; these operations induce spikes in the input bias current that may be converted into voltage spikes in high source impedance applications or during use of high-value resistors to gain up the signal. For this reason, in applications with a narrow temperature range such as medical instrumentation, the near-zero drift may not be as important as the ultra-low input bias current of the amplifier coupled with high source impedance. In such cases, the OPA2191 with 20-pA (max) bias current is likely a better option than the OPA2187 with much lower offset drift but 350-pA (max) bias current.

Summary

Table 1 compares the three offset correction methods. This technical brief may serve as a guide to help you select an amplifier that best meets your low-offset, low-drift circuit needs.

Table 1. Laser Trim, e-Trim, and Chopper Comparison

Devices	Correction Method	Advantages	Disadvantages
INA826S OPA145 OPA2325	Laser trim	Simplified manufacturing flow over temperature	Parameter shifts after packaging
OPA2191 OPA2192 OPA2376	e-Trim	No parameter shifts because of packaging	Complex manufacturing flow over temperature
INA188 OPA189 OPA2187 OPA388	Chopper	Lowest possible drift; no 1/f noise	Bias current spikes can cause problems with high source impedances

Table 2. Related Blog Posts and Resources

Video	Op Amp Technology Overview
Precision Hub Blog	How does package level trim compare to other offset correction methods?
Precision Hub Blog	Trimmed or chopped: How do you like your op amp?
EDN Analog Design Post	Pushing the precision envelope – Understanding the precision challenge in operational amplifiers

Copyright © 2017, Texas Instruments Incorporated

IMPORTANT NOTICE FOR TI DESIGN INFORMATION AND RESOURCES

Texas Instruments Incorporated ('TI") technical, application or other design advice, services or information, including, but not limited to, reference designs and materials relating to evaluation modules, (collectively, "TI Resources") are intended to assist designers who are developing applications that incorporate TI products; by downloading, accessing or using any particular TI Resource in any way, you (individually or, if you are acting on behalf of a company, your company) agree to use it solely for this purpose and subject to the terms of this Notice.

TI's provision of TI Resources does not expand or otherwise alter TI's applicable published warranties or warranty disclaimers for TI products, and no additional obligations or liabilities arise from TI providing such TI Resources. TI reserves the right to make corrections, enhancements, improvements and other changes to its TI Resources.

You understand and agree that you remain responsible for using your independent analysis, evaluation and judgment in designing your applications and that you have full and exclusive responsibility to assure the safety of your applications and compliance of your applications (and of all TI products used in or for your applications) with all applicable regulations, laws and other applicable requirements. You represent that, with respect to your applications, you have all the necessary expertise to create and implement safeguards that (1) anticipate dangerous consequences of failures, (2) monitor failures and their consequences, and (3) lessen the likelihood of failures that might cause harm and take appropriate actions. You agree that prior to using or distributing any applications that include TI products, you will thoroughly test such applications and the functionality of such TI products as used in such applications. TI has not conducted any testing other than that specifically described in the published documentation for a particular TI Resource.

You are authorized to use, copy and modify any individual TI Resource only in connection with the development of applications that include the TI product(s) identified in such TI Resource. NO OTHER LICENSE, EXPRESS OR IMPLIED, BY ESTOPPEL OR OTHERWISE TO ANY OTHER TI INTELLECTUAL PROPERTY RIGHT, AND NO LICENSE TO ANY TECHNOLOGY OR INTELLECTUAL PROPERTY RIGHT OF TI OR ANY THIRD PARTY IS GRANTED HEREIN, including but not limited to any patent right, copyright, mask work right, or other intellectual property right relating to any combination, machine, or process in which TI products or services are used. Information regarding or referencing third-party products or services does not constitute a license to use such products or services, or a warranty or endorsement thereof. Use of TI Resources may require a license from a third party under the patents or other intellectual property of the third party, or a license from TI under the patents or other intellectual property of TI.

TI RESOURCES ARE PROVIDED "AS IS" AND WITH ALL FAULTS. TI DISCLAIMS ALL OTHER WARRANTIES OR REPRESENTATIONS, EXPRESS OR IMPLIED, REGARDING TI RESOURCES OR USE THEREOF, INCLUDING BUT NOT LIMITED TO ACCURACY OR COMPLETENESS, TITLE, ANY EPIDEMIC FAILURE WARRANTY AND ANY IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, AND NON-INFRINGEMENT OF ANY THIRD PARTY INTELLECTUAL PROPERTY RIGHTS.

TI SHALL NOT BE LIABLE FOR AND SHALL NOT DEFEND OR INDEMNIFY YOU AGAINST ANY CLAIM, INCLUDING BUT NOT LIMITED TO ANY INFRINGEMENT CLAIM THAT RELATES TO OR IS BASED ON ANY COMBINATION OF PRODUCTS EVEN IF DESCRIBED IN TI RESOURCES OR OTHERWISE. IN NO EVENT SHALL TI BE LIABLE FOR ANY ACTUAL, DIRECT, SPECIAL, COLLATERAL, INDIRECT, PUNITIVE, INCIDENTAL, CONSEQUENTIAL OR EXEMPLARY DAMAGES IN CONNECTION WITH OR ARISING OUT OF TI RESOURCES OR USE THEREOF, AND REGARDLESS OF WHETHER TI HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.

You agree to fully indemnify TI and its representatives against any damages, costs, losses, and/or liabilities arising out of your non-compliance with the terms and provisions of this Notice.

This Notice applies to TI Resources. Additional terms apply to the use and purchase of certain types of materials, TI products and services. These include; without limitation, TI's standard terms for semiconductor products http://www.ti.com/sc/docs/stdterms.htm), evaluation modules, and samples (http://www.ti.com/sc/docs/sampterms.htm).

Mailing Address: Texas Instruments, Post Office Box 655303, Dallas, Texas 75265
Copyright © 2017, Texas Instruments Incorporated